

Springboard Virtual Internship 6.0

Name: SIVARAJ S

Domain: Data Visualization

Project: Visualizing U.S. Natural Disaster Declarations

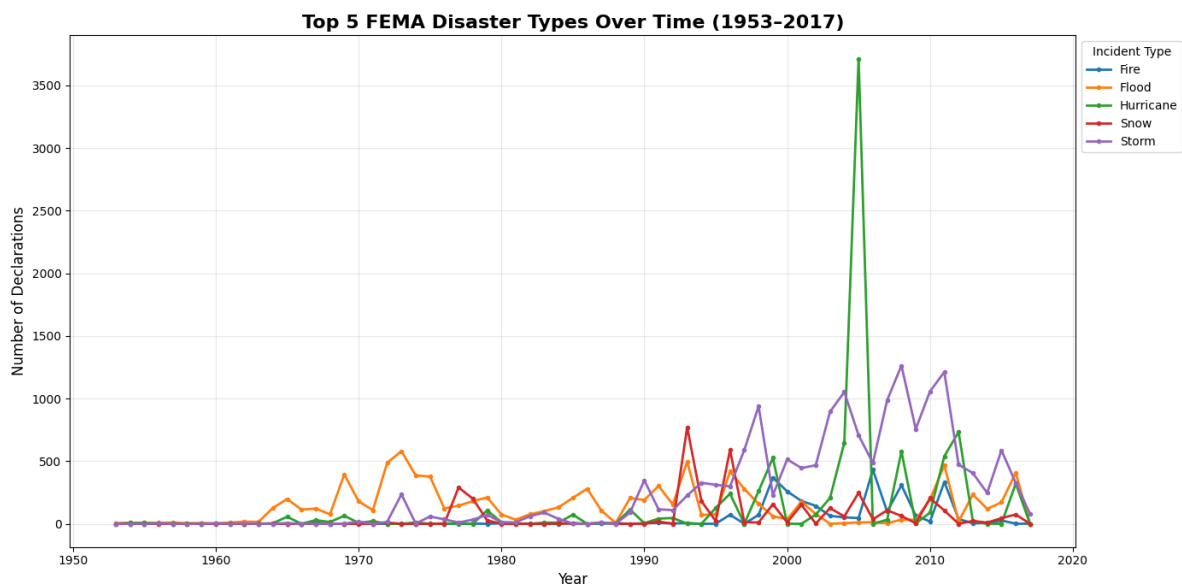
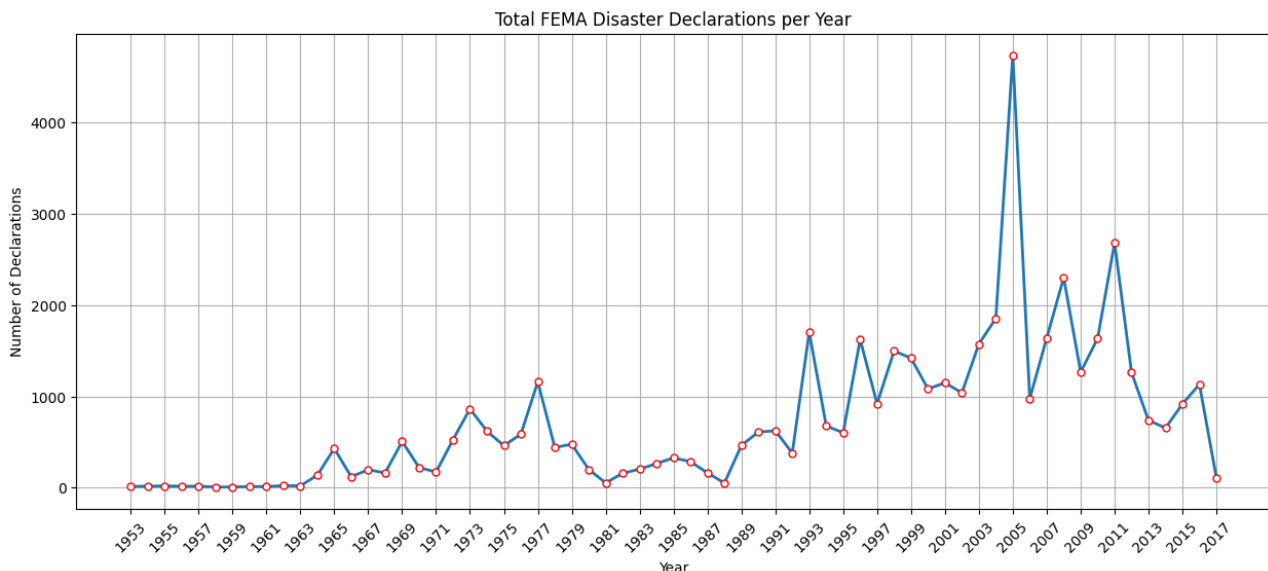
Mentor: G.K.S Jyoteesh

INSIGHTS REPORT

1. Overview of Insights

- This report summarizes the key insights derived from the analysis of U.S. natural disaster declarations.
- The insights are based on temporal, geographic, and incident type analysis performed across all milestones.
- The objective of this report is to identify patterns, trends, and relationships in disaster occurrences and provide a clear understanding of disaster behaviour over time and across regions.

Yearly Trend Analysis:

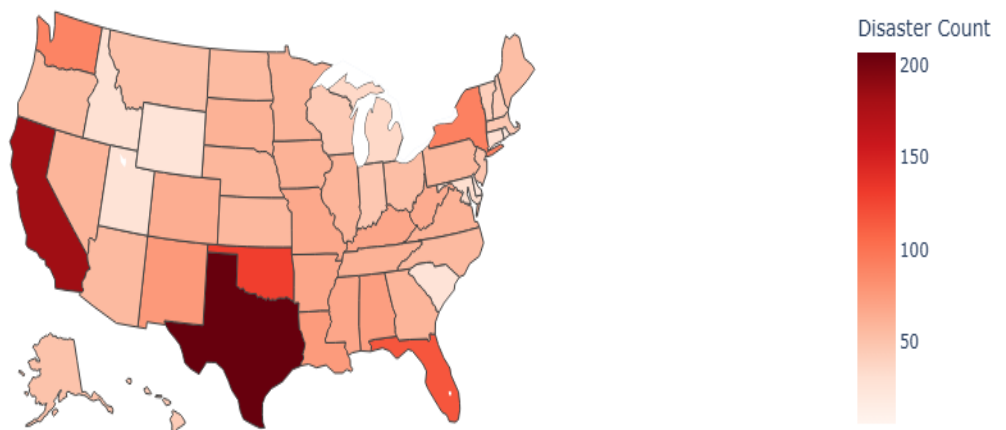


- The graph shows the total number of FEMA disaster declarations per year from 1953 to 2017.
- There is an overall increasing trend in disaster declarations, with a major spike around 2005.
- The increase may be due to more extreme weather events, climate change, and improved disaster reporting systems

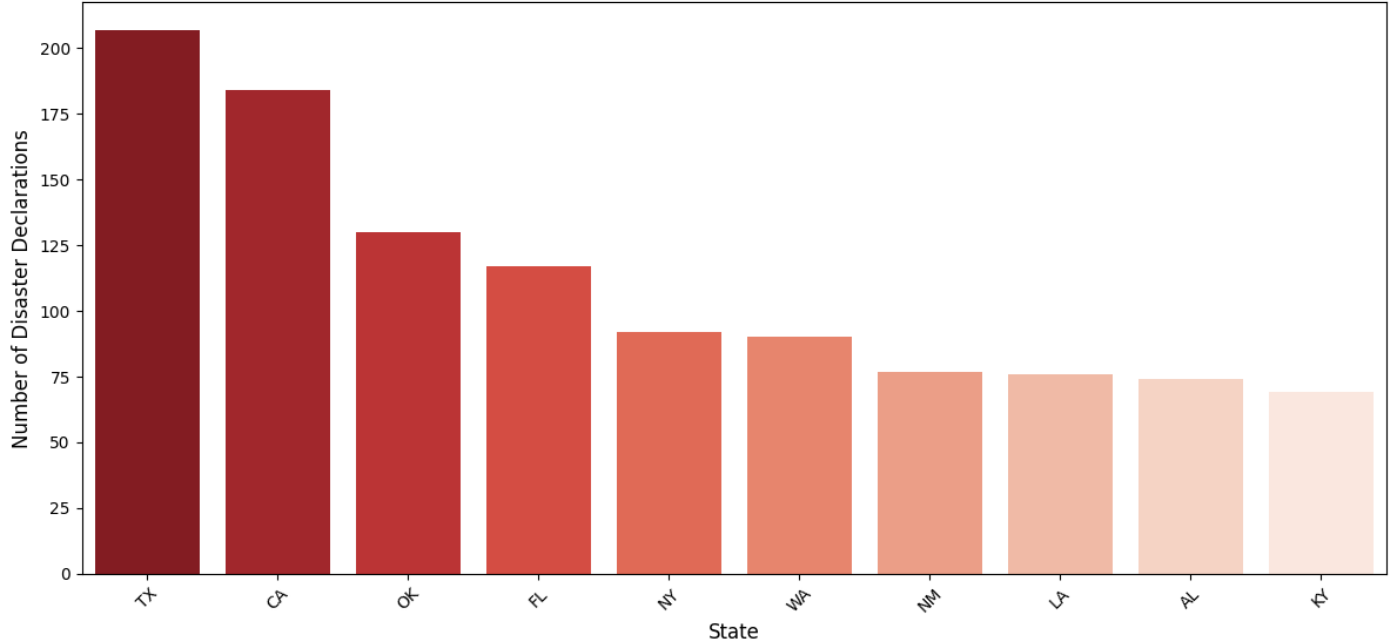
The analysis of FEMA disaster declarations reveals a clear increasing trend in disaster occurrences over time. The visualizations show that certain disaster types such as storms, floods, and hurricanes occur more frequently, and disasters also display seasonal patterns during specific months. Additionally, some years show significant spikes in disaster declarations, indicating periods of extreme disaster activity. Overall, the analysis highlights important temporal trends and patterns in disaster events, demonstrating how data visualization can help better understand historical disaster behaviour.

3.Geographic analysis:

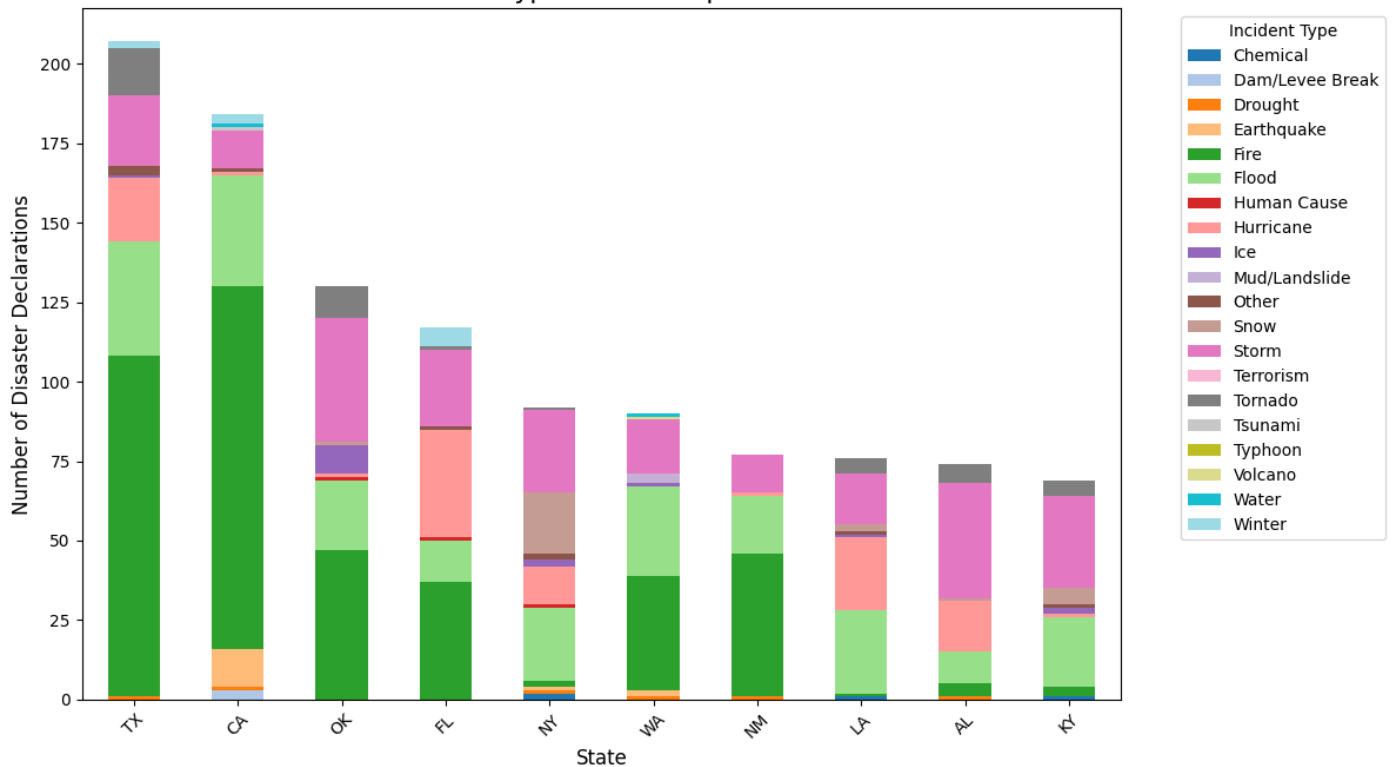
Geographical Distribution of Natural Disaster Declarations Across the United States



Top 10 States with Highest Natural Disaster Declarations



Distribution of Disaster Types Across Top Disaster-Prone States



Geographic trend analysis shows that disaster declarations are concentrated in a few high-risk states rather than spread evenly across the country. Texas, California, and Florida consistently appear as the most affected states, which indicates strong regional vulnerability linked to climate and geography.

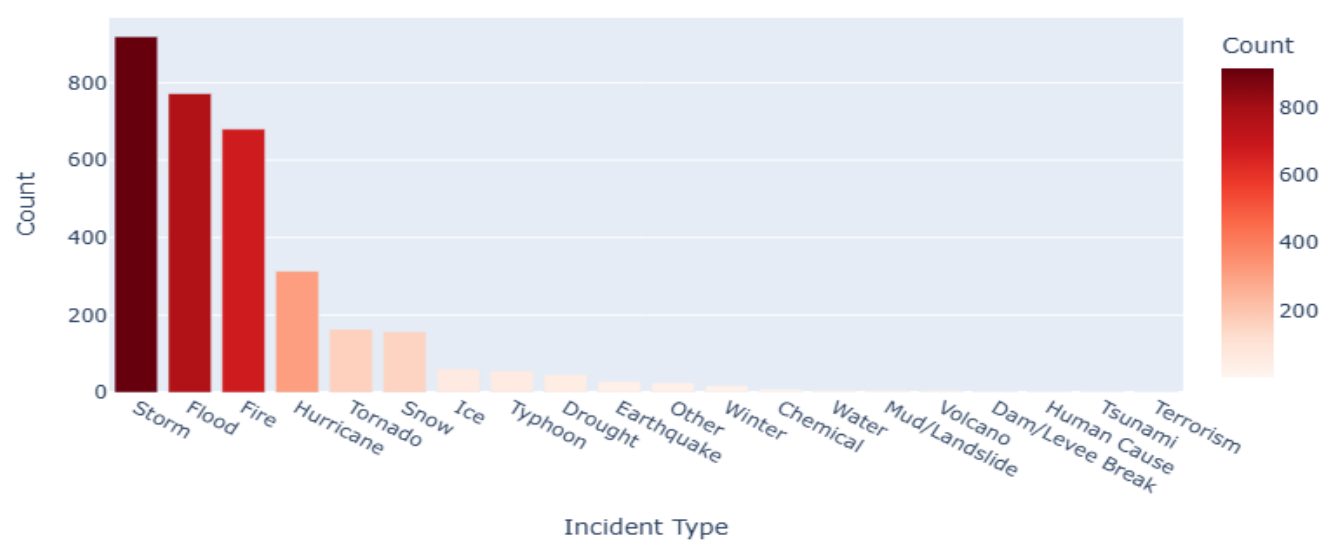
The state-by-incident breakdown also shows clear regional disaster patterns. Coastal and southern states are more exposed to hurricanes and floods, western states face more fire-

related disasters, and central states tend to experience more severe storms and tornadoes. This means each region has its own dominant disaster profile.

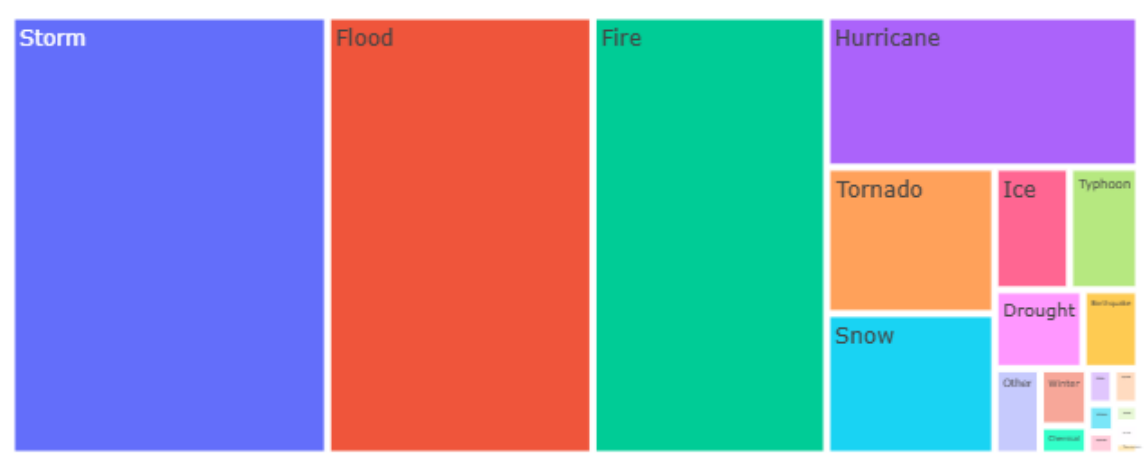
Overall, the geographic trend analysis confirms that disaster risk is strongly location-based, with repeated hotspots in specific states. These patterns suggest that preparedness and response planning should be tailored to each region’s main disaster types rather than using a one-size-fits-all approach.

Incident Type Analysis:

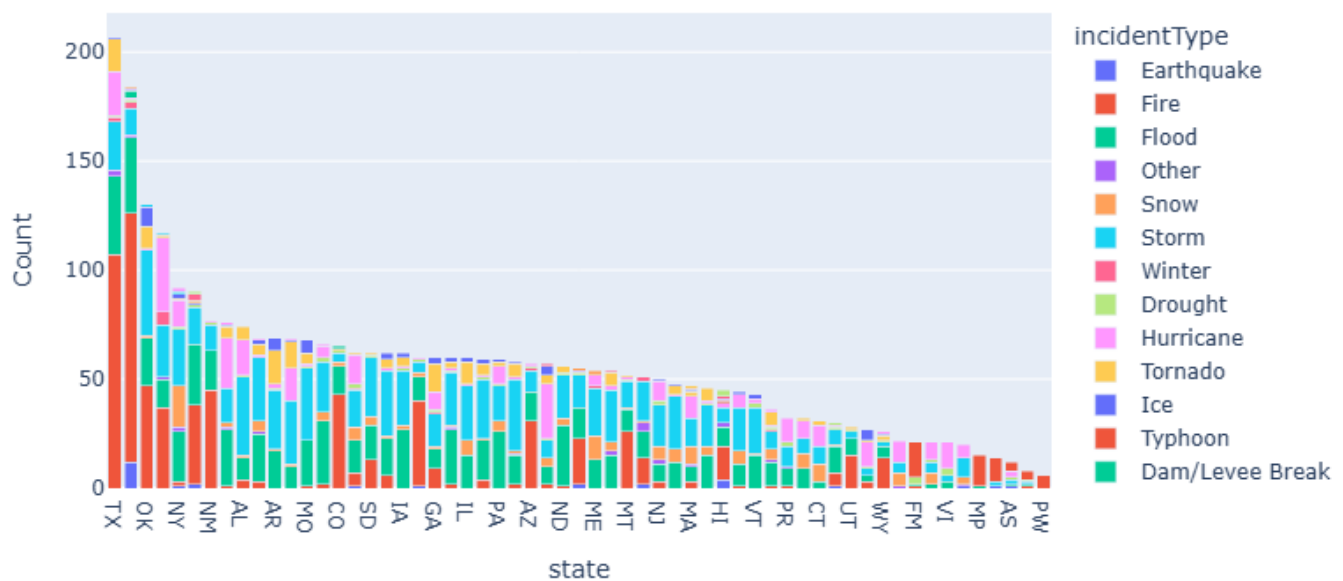
Most Frequent Disaster Types



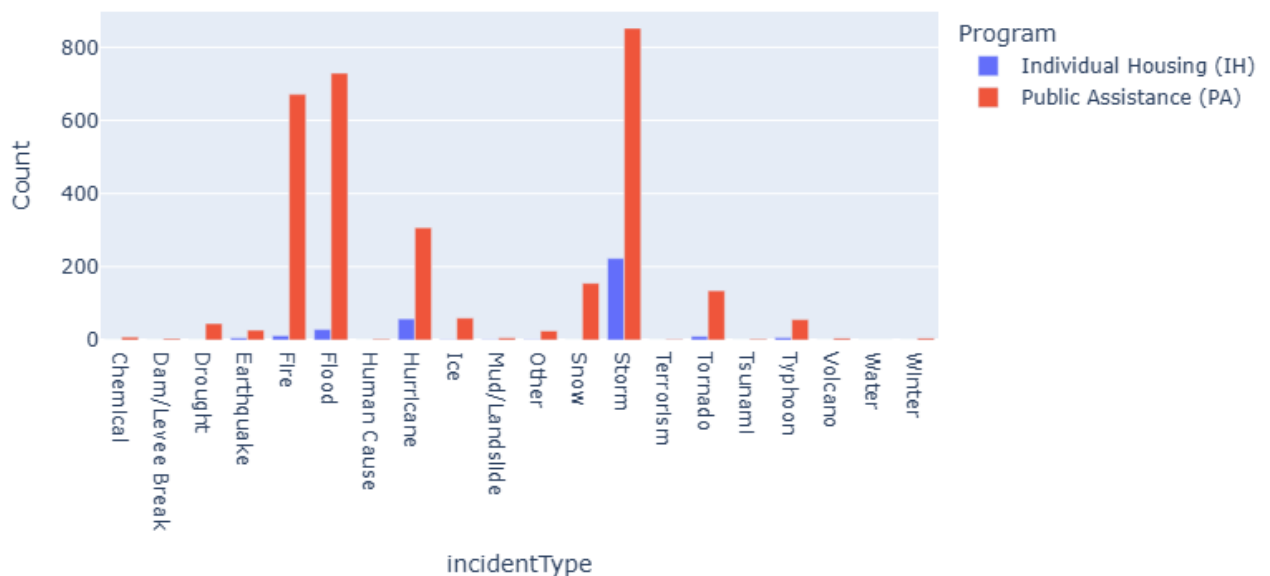
Disaster Composition (Treemap)



Incident Type Contribution within Each State



Disaster Assistance Programs by Incident Type



incident type analysis shows that a small number of disaster types account for most declarations, with floods, severe storms, and hurricanes appearing as the most frequent categories overall. The state-by-incident pattern also shows strong regional differences: coastal and southern states are more affected by hurricanes and floods, while western states see more fire-related incidents.

The assistance program analysis adds another layer of insight. Public Assistance is activated more broadly across major disaster types, while Individual Housing support is more concentrated in severe events such as hurricanes and widespread flooding. Overall, the milestone confirms that disaster type, geography, and federal aid patterns are closely linked.

FINAL INSIGHT:

The overall project shows that natural disaster declarations in the U.S. are driven by strong time, place, and incident-type patterns rather than random occurrence. Over time, disaster frequency has generally increased; geographically, risk is concentrated in a few high-impact states like Texas, California, and Florida; and by incident type, floods, storms, hurricanes, and fires dominate the dataset depending on region.

The final takeaway is that disaster preparedness cannot be uniform. Coastal and southern states need stronger hurricane and flood response, western states need wildfire-focused planning, and central states need storm and tornado readiness. The assistance program analysis also shows that major disasters often trigger broad federal support, especially Public Assistance, with Individual Housing aid concentrated in the most severe events.

DEPLOYED DASHBOARD LINK:

LINK : <https://usnd-iota.vercel.app/>